CIVIC PULSE / PUBLIC INFRASTRUCTURE

Civic Triage

Updated 40-Hour GenAI Hackathon Roadmap

A citizen reports a local issue. AI observes the evidence. Deterministic backend rules set priority and department. Staff can override decisions, and a public dashboard tracks the ticket.

THE BUILD TARGET

Ship one trustworthy end-to-end loop: photo / optional voice / GPS → multimodal observations → transparent triage rules → staff review and override → public accountability dashboard.

What we are building

A phone-friendly civic reporting web app for four issue categories: pothole, water leak, garbage accumulation and broken streetlight. Reports include an image, optional typed or spoken description, and GPS coordinates. The system creates a ticket, recommends a responsible department, flags uncertain or high-risk cases for human review, and shows status on a map.

Scope boundary

Build in 40 hours	Do not build during the hackathon
Image + GPS report form; optional voice transcription	Custom model fine-tuning or training a new detector
Gemini / Groq vision triage with validated JSON	Full social-media monitoring or autonomous government portal access
Whisper transcription; optional AST sound tags	Complex multi-agent orchestration
Backend severity rules + department routing	DistilBERT text router or AST fine-tuning
Public map, staff queue, status flow, override + audit log	Real municipal integration without authorization

NON-NEGOTIABLE DESIGN RULE

The model perceives; the backend decides; a human can override. Never present model-generated confidence as a calibrated probability.

Team assumption: 3–4 people. Duration: 40 hours. Target: a deployable demo with one honest evaluation result, not a production municipal system.

1. Model and API plan

Use hosted APIs for the application's main inference path. Use Colab only for local model experiments and evaluation. Model identifiers below were checked against provider documentation on 31 October 2026; re-check access and rate limits before the event.

Task	Primary	Fallback / experiment	Where it runs
Image understanding / civic issue triage	Gemini 3.8 Flash `gemini-3.8-flash`	Groq Qwen 3.8 27B Vision `qwen/qwen3.8-27b`	Gemini / Groq API from FastAPI
Local VLM benchmark	-	Qwen2-VL-7B-Instruct `Qwen/Qwen2-VL-7B-Instruct`	Colab GPU, 4-bit experiment only
Speech-to-text	Groq Whisper Large V3 Turbo `whisper-large-v3-turbo`	Gemini 3.5 Transcribe `gemini-3.5-transcribe`	Groq API; Gemini optional fallback
Environmental audio tags	-	MIT AST AudioSet `MIT/ast-finetuned-audioset-10-10-0.4593`	Colab GPU / local Python; optional
Final severity / department / review	Python deterministic rules	Staff override	FastAPI backend; never delegated to LLM

Recommended provider routing

- Vision primary: Gemini 3.8 Flash. It accepts image input and supports schema-constrained JSON output. Use it for issue type, visible evidence, image quality, hazards and suggested severity.
- Vision fallback: Groq-hosted Qwen 3.8 27B (`qwen/qwen3.8-27b`). It accepts image + text and supports JSON mode. Groq currently labels this model Preview, so verify account access and availability before the event. Invoke only when Gemini is unavailable or fails validation after one retry.
- Audio primary: Groq Whisper Large V3 Turbo (`whisper-large-v3-turbo`) for fast multilingual transcription. Use Whisper Large V3 instead if transcription accuracy matters more than speed/cost.
- Audio fallback: Gemini 3.5 Transcribe (`gemini-3.5-transcribe`) through the Gemini Interactions API. Keep this adapter separate because its request flow differs from Groq's OpenAI-compatible audio endpoint.
- AST extension: use MIT AST to add AudioSet sound-event tags only when a voice/audio note is available and core work is stable. Treat tags as supporting audio observations, not proof of a leak or other civic issue.

> **IMPORTANT HF TOKEN DISTINCTION**
>
> HF_TOKEN lets Colab download public model weights and use gated models where access is granted. It does not automatically give hosted inference for Qwen2-VL-7B or MIT AST. Both model pages currently indicate no Inference Provider deployment; run them in Colab/local Transformers or deploy your own endpoint.

What to benchmark before choosing the default

Run the same 30 labelled civic images through Gemini and Groq Qwen vision. Compare issue-type accuracy, severity within ±1 of the reviewer label, malformed-output rate and latency. Keep Gemini as default unless the measured test shows a clear reason to switch. Do not call both providers for every public report.

2. System architecture

END-TO-END REQUEST PATH

React report form → FastAPI upload endpoint → parallel / sequential model adapters (Whisper for audio, Gemini for image; Groq Qwen fallback) → Pydantic validation → deterministic decision engine → SQLite ticket + files → Leaflet public map and staff queue.

Layer	Responsibility	Implementation
Citizen app	Capture photo, typed description, optional voice, GPS and consent	React + Vite + Tailwind; browser camera / geolocation APIs
API and file handling	Validate type/size, store evidence, create ticket request	FastAPI; local uploads directory; MIME checks
Audio adapter	Transcribe citizen speech; retain transcript as citizen-reported text	Groq Whisper API; Gemini Transcribe fallback
Vision adapter	Describe only what is visible; return strict observation schema	Gemini 3.8 Flash; Groq Qwen 3.8 27B fallback
Optional AST adapter	Add broad environmental sound tags from audio clip	MIT AST AudioSet in Colab/local Python
Decision engine	Apply severity rubric, department map and review flags	Pure Python functions + unit tests
Persistence	Tickets, provider traces, status history and staff audit events	SQLite + local image/audio storage
Dashboard	Map, ticket list/detail, staff queue, analytics and overrides	React + Leaflet; charts optional

Keep evidence provenance separate

Data layer	Source	UI label
Visible in photo	Vision model `visual_evidence`	AI observed in photo
Citizen-reported	Typed text or Whisper transcript	Citizen says (unverified report)
Audio event tags (optional)	AST AudioSet predictions	Audio model detected (supporting signal)
AI inference	Suggested severity, hazards, justification	AI suggestion - not final decision
Final decision	Backend rules + human override	Final ticket decision

Backend API endpoints

Method + route	Purpose
POST `/api/reports`	Create report: image, text/audio, latitude, longitude
GET `/api/tickets`	Public ticket list with filters and pagination
GET `/api/tickets/{id}`	Ticket detail, provenance, evidence and status history
GET `/api/analytics/summary`	Non-simulated operational statistics
GET `/api/staff/review-queue`	Tickets requiring human review
PATCH `/api/staff/tickets/{id}`	Assign, change status or override type/severity/department with reason
GET `/api/health/providers`	Report provider connectivity without exposing secrets

Do not call model APIs from React. Keep all provider keys and file-processing logic in FastAPI.

3. Model contract and decision rules

Agree on the contract by hour 1. The VLM must return observations only. It must not choose the final department, final severity, ticket status or human-review decision.

Vision observation JSON

```
{
  "issue_type": "pothole | water_leak | garbage | streetlight | other | not_civic",
  "visual_evidence": ["Large road depression", "Standing water inside it"],
  "suggested_severity": 4,
  "justification": "Visible damage may pose a risk to vehicles.",
  "hazards": ["possible vehicle damage"],
  "image_quality": "good | poor | unusable",
  "consistent_with_citizen_report": "yes | no | unclear | no_report",
  "model_confidence_uncalibrated": 0.8
}
```

If the transcript is passed to the vision model, label it explicitly as an unverified citizen statement. Use it only for the consistency field; never copy spoken claims into `visual_evidence`.

Audio record (kept separate)

```
{
  "transcript": "There is water leaking beside the road.",
  "transcript_language": "hi",
  "asr_provider": "groq",
  "asr_model": "whisper-large-v3-turbo",
  "ast_tags": ["Water", "Traffic"]
}
```

`ast_tags` is optional and should be omitted if AST is not integrated. AudioSet labels are broad sound classes; they are not a trained civic complaint classifier.

Backend severity rubric (1–5)

1. Start from `suggested_severity` after type/range validation.
2. If image quality is poor, cap severity at 3.
3. If a hazard indicates an open manhole, exposed wire, live current or flooding, floor severity at 4. Safety floor overrides the poor-image cap.
4. If issue is `water_leak` and a hazard mentions road or electricity, add 1.
5. Clamp final severity to 1–5. Store `rules_applied` so the displayed decision is explainable.

Department mapping

Issue type	Department
pothole	Roads
water_leak	Water
garbage	Sanitation
streetlight	Electrical
other / not_civic	Triage desk

Set `needs_human_review = true` when

- Model JSON is invalid after one retry, issue type is `other` or `not_civic`, image quality is poor/unusable, or visual evidence is empty.
- Final severity is 4 or 5, citizen statement conflicts with the image, or GPS is missing.
- Staff queue should show the review reason. Public map status should say “Awaiting review” until a person validates the ticket.

> **CONFIDENCE POLICY**
>
> Store `model_confidence_uncalibrated` for debugging only. Never display it as “92% sure”. Human review is determined by explicit rules, not by the model’s self-reported confidence.

4. What runs in Colab vs. the application

Environment	Run here	Do not run here
Google Colab	Qwen2-VL-7B 4-bit benchmark; MIT AST inference; sample Whisper/Gemini/Groq API smoke tests; evaluation notebook	React app, persistent FastAPI backend, SQLite production data, public dashboard
Local VS Code	FastAPI, decision rules, Pydantic schemas, SQLite, upload storage, React/Vite, Leaflet, provider adapters	Long-lived GPU model serving unless local GPU is deliberately chosen
Hosted APIs	Gemini image reasoning; Groq Qwen vision fallback; Groq Whisper transcription; optional Gemini transcription fallback	Local model fine-tuning or custom AST classification

Colab notebook A: VLM benchmark

- Install Transformers, Accelerate, BitsAndBytes and qwen-vl-utils; check the allocated GPU and available VRAM.
- Authenticate with HF_TOKEN if required; load `Qwen/Qwen2-VL-7B-Instruct` in 4-bit. This is an experiment, not a runtime dependency.
- Run 30–40 civic photos; save raw output, parsed JSON, latency and errors to CSV/JSON.
- Compare the outputs against the Gemini and Groq API responses on the same labelled set.

Colab notebook B: AST sound-event experiment

- Load `MIT/ast-finetuned-audioset-10-10-0.4593` with Transformers `pipeline("audio-classification", ...)` or AutoFeatureExtractor + AutoModelForAudioClassification.
- Use the model's expected audio preprocessing/sample rate; test short clips for water, traffic, machinery and street ambience.
- Record top-k AudioSet tags and inspect whether they help. Do not make AST required for ticket creation.
- The model is AudioSet-pretrained and predicts AudioSet classes; it has no hosted Inference Provider at the time of verification.

Colab notebook C: API smoke tests

- Test Gemini image input with a sample pothole/garbage/leak/streetlight photo and request schema-shaped JSON.
- Test Groq Qwen vision using image URL or base64 image; test JSON mode.
- Test Groq Whisper Large V3 Turbo with Hindi and English voice notes; compare one small sample with Gemini 3.5 Transcribe.

> **COLAB WARNING**
>
> Do not expose a Colab notebook as your production API. Runtime disconnects and temporary storage make it unsuitable for the live demo. Use Colab to test and benchmark; call stable provider APIs from FastAPI.

Environment variables

```
HF_TOKEN=hf_...                  # optional for model downloads / gated access
GROQ_API_KEY=gsk_...
GEMINI_API_KEY=...

GEMINI_VISION_MODEL=gemini-3.8-flash
GROQ_VISION_MODEL=qwen/qwen3.8-27b
GROQ_ASR_MODEL=whisper-large-v3-turbo
GEMINI_TRANSCRIBE_MODEL=gemini-3.5-transcribe
HF_VISION_MODEL=Qwen/Qwen2-VL-7B-Instruct
AST_MODEL=MIT/ast-finetuned-audioset-10-10-0.4593
```

Store secrets only in backend environment / secret manager / Colab Secrets. Never put them in Vite variables (`VITE_*`) or commit `.env` to Git.

5. Updated 40-hour schedule

Keep the original six delivery gates: scope/schema frozen (H3), model outputs valid (H9), first end-to-end ticket (H15), staff override works (H24), feature freeze (H30), code freeze (H35).

Time	Workstream / task	Exit condition
H0–3	Lock issue types, 1–5 severity rubric, department map, output contract, provider choice and team ownership. Create repo and task board.	Schema + rubric frozen
H3–9	AI: Gemini image call, Groq Qwen fallback, Groq Whisper smoke test; prompt + Pydantic parse. Frontend: phone-friendly form + GPS. Backend: FastAPI scaffold + upload endpoint + SQLite.	Valid JSON on test images; sample audio transcribed
H9–15	Wire report → stored media → model inference → decision rules → ticket saved. Add retry once and human-review fallback.	First real end-to-end ticket
H15–24	Build public map, list filters, ticket detail, staff review queue, status workflow, staff override with reason and audit log.	Staff override works end-to-end
H24–30	Build summary metrics + methodology page; seed 40–60 demo tickets with `is_simulated=true`; prepare 30-image labelled evaluation set. AST experiment only if ahead.	Feature freeze at H30
H30–35	Run end-to-end QA, test poor/irrelevant photos, silent audio, missing GPS, provider timeout and invalid JSON. Measure model/reviewer agreement.	Code freeze at H35
H35–38	Deploy frontend/backend, configure production secrets, verify file upload and provider connectivity, record backup demo video.	Live link + backup video
H38–40	Prepare five-slide deck and rehearse a three-minute demo. No new features.	Pitch ready

Parallel team split (3–4 people)

Owner	Primary tasks
AI / model engineer	Gemini & Groq vision adapters, Whisper adapter, prompt, provider fallback, optional AST + benchmark
Backend engineer	FastAPI, Pydantic, SQLite, media storage, decision engine, status transitions and audit log
Frontend engineer	React form, camera/mic/GPS, Leaflet map, ticket detail, staff queue and charts
Integration / QA / demo	Seed data, 30 labelled images, test cases, deployment, backup video and pitch; pair on integration

PARALLELISM RULE

Freeze the JSON contract first. Frontend can use mock responses while AI and backend work independently. Integrate by H9 and protect the H15 end-to-end milestone.

6. Evaluation, reliability and trust

Evaluation set

Hand-label 30 images by hour 33 (or earlier if possible) with issue type, reviewer severity 1–5 and whether human review is required. Use the same images to compare Gemini and Groq Qwen vision; run Qwen2-VL in Colab as an optional third benchmark.

Metric	How to report it
Issue-type accuracy	Correct issue type / 30. Include sample size.
Severity agreement	Share of cases where model suggestion is within ±1 of reviewer label.
High-severity review coverage	Share of final-severity 4–5 tickets sent to human review. Target 100% under the rule set.
Structured-output reliability	Valid schema responses on first attempt; separately report retry/fallback count.
Latency	Median and p95 end-to-end inference time on the test set, if measured.

This is a small sanity check, not a benchmark. Do not claim broad accuracy from 30 images.

Required failure tests

- Blurry photo / unusable image → human review; do not infer a confident severity.
- Irrelevant photo → `not_civic` and triage desk / manual review.
- Silent or corrupt audio → keep the image report; mark transcription unavailable.
- Missing GPS → create ticket only if product policy allows, but flag for review and do not pretend the location is known.
- Gemini timeout / quota error → one retry with backoff, then Groq Qwen vision. If both fail, ticket is saved as awaiting review.
- Malformed JSON → validate with Pydantic, retry once, then human review; never infer a department from malformed model text.

Trust and privacy rules

- Show three provenance labels: “AI observed in photo”, “Citizen says”, and “AI suggestion”. Keep AST tags in a fourth optional audio-evidence field.
- Staff can override issue type, severity or department only with a reason. Record old value, new value, actor, timestamp and reason in an audit log.
- Demo tickets carry `is_simulated=true`, a visible “Demo data” banner and are excluded from real metrics.
- Do not expose citizen names, phone numbers or raw voice recordings publicly. Keep precise coordinates for staff workflow; consider approximate public map pins.
- Set file size/type limits, randomize stored filenames, validate MIME type, and do not log API keys or raw secrets.

Cut order if behind schedule

1. AST sound tagging / any optional model benchmarking.
2. Face and number-plate blurring.
3. Geo-radius duplicate count.
4. Reduce seeded demo tickets from 60 to 30.
5. Drop voice capture and keep typed text only; keep image upload and triage.

> **NEVER CUT**
>
> Human-review flags, staff override with audit log, separate provenance labels, backend decision rules and “Demo data” labels.

7. Demo, deployment and references

Three-minute demo script

1. Citizen opens the phone-friendly form, photographs a pothole, adds a short spoken or typed note, and shares GPS.
2. Show the transcript under “Citizen says” and the image observations under “AI observed in photo”.
3. Show suggested severity and backend-applied rules; the final ticket routes to Roads and appears on the map.
4. Staff opens the review queue, changes severity or department with a reason, and saves the override. Show the audit entry.
5. Move the ticket through Assigned → In progress → Resolved. Show the public accountability dashboard and visible Demo data label.
6. Close with one honest 30-image metric, the scope limitations and next steps.

Deployment checklist

- Frontend on Vercel or equivalent static hosting; backend on Render or another persistent Python host.
- Configure GEMINI_API_KEY and GROQ_API_KEY as backend secrets; HF_TOKEN only where Colab/model download needs it.
- Use persistent storage or managed database for deployed tickets; local SQLite and local file storage are acceptable for local development but may not persist on some free hosting plans.
- Test upload size, CORS, health check, model-provider errors, map tiles, and one full report after deployment.
- Record a backup demo video before the final hour; keep 3–5 known-good report examples ready.

Official model and API references

Gemini API models and IDs https://ai.google.dev/gemini-api/docs/models
Gemini image understanding / image input https://ai.google.dev/gemini-api/docs/image-understanding
Gemini structured outputs https://ai.google.dev/gemini-api/docs/structured-output
Gemini 3.5 Transcribe https://ai.google.dev/gemini-api/docs/transcribe
Groq vision and Qwen 3.8 27B https://console.groq.com/docs/vision
Groq speech-to-text / Whisper models https://console.groq.com/docs/speech-to-text
Groq active models and limits https://console.groq.com/docs/models
Qwen2-VL-7B-Instruct model card https://huggingface.co/Qwen/Qwen2-VL-7B-Instruct
MIT AST AudioSet model card https://huggingface.co/MIT/ast-finetuned-audioset-10-10-0.4593
FastAPI documentation https://fastapi.tiangolo.com/
Leaflet documentation https://leafletjs.com/reference.html

Final build principle

> **ONE TRUSTWORTHY LOOP**
>
> A useful civic product does not need the most models. It needs good evidence handling, predictable decisions, transparent routing, human oversight, a working public status flow and honest evaluation.

Prepared 31 October 2026. Model identifiers and provider documentation were checked during preparation; confirm model availability, account access, quotas and pricing in the provider consoles immediately before the hackathon.